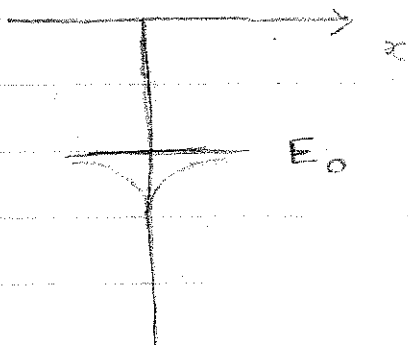


Homework Set #6

S5.2.3

$$V(x) = -aV_0 \delta(x)$$



$$H|E\rangle = +E|E\rangle \quad (E < 0)$$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \psi_E(x) = E \psi_E(x)$$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_E(x) = E \psi_E(x) \quad x \neq 0$$

$$-\frac{\hbar^2}{2m} \left[\frac{d\psi_E(0_+)}{dx} - \frac{d\psi_E(0_-)}{dx} \right]$$

$$= \lim_{E \rightarrow 0} \int_{-E}^{+E} dx [E - V(x)] \psi_E(x)$$

$$= +aV_0 \psi_E(0)$$

$$\psi_E(x) = A_+ e^{-\kappa x} + B_+ e^{\kappa x} \quad x > 0$$

$$\text{with } \kappa = \frac{1}{\hbar} \sqrt{-2mE}$$

$$\psi_E(x) = A_- e^{-\kappa x} + B_- e^{\kappa x} \quad x < 0$$

The wavefunction must be normalizable

$$B_+ = A_- = 0$$

As was shown in class, the wavefunction must be continuous

$$\therefore A_+ = B_-$$

Hence

$$\psi_E(x) = A e^{-K|x|}$$

$$-\frac{\hbar^2}{2m} [-AK - (AK)] = aV_0 A$$

$$\frac{\hbar^2}{m} K = aV_0$$

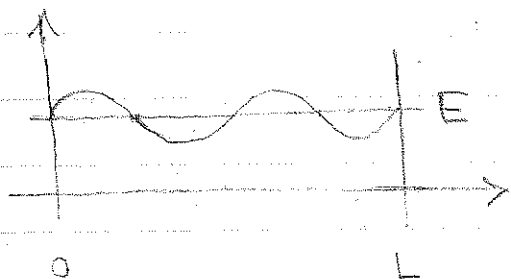
$$K = \frac{maV_0}{\hbar^2}$$

$$E_0 = -\frac{\frac{\hbar^2}{m} K^2}{2m} = -\frac{ma^2 V_0^2}{2\hbar^2}$$

There are no other bound states, since our method would have found them if they existed.

S5.2.4

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$



$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L}\right)^2$$

$$F_n = -\frac{\partial E_n}{\partial L} = \frac{\hbar^2 n^2 \pi^2}{m L^3} = \frac{2E_n}{L}$$

A classical particle of energy E_n has magnitude of momentum

$$|p_n| = \sqrt{2mE_n} \quad \text{It hits the}$$

RHS wall at time intervals

$$\Delta t = \frac{2L}{|v_n|} = \frac{2Lm}{|p_n|} = L \sqrt{\frac{2m}{E_n}}$$

Each time it hits that wall it

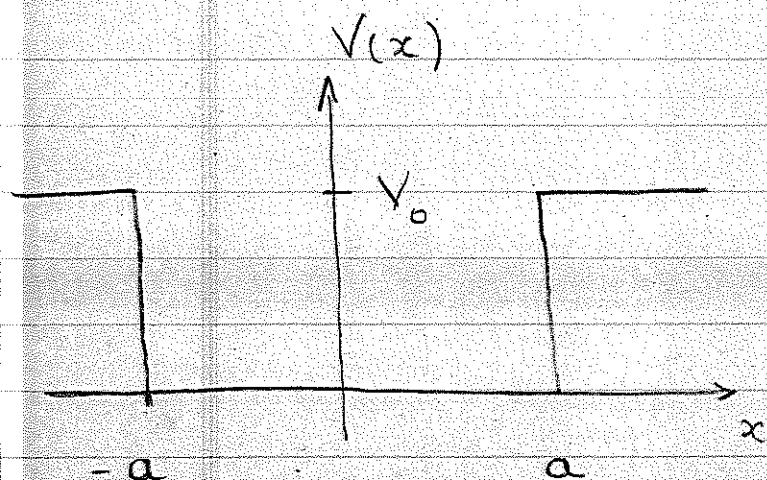
changes its momentum by $2|p_n|$.

Thus

$$F_n^{\text{class}} = \frac{2|p_n|}{\Delta t} = \frac{2|p_n|^2}{2Lm} = \frac{2E_n}{L}$$

which equals F_n .

S 5.2.6



$$\frac{-\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = [E - V(x)] \psi$$

$$E < V_0$$

$$i. \quad \begin{cases} \frac{d^2 \psi}{dx^2} = -k^2 \psi, & k \equiv \frac{1}{\hbar} \sqrt{2mE} \quad \text{for } |x| < a \\ \frac{d^2 \psi}{dx^2} = \kappa^2 \psi, & \kappa \equiv \frac{1}{\hbar} \sqrt{2m(V_0 - E)} \quad \text{for } |x| > a \end{cases}$$

$$\begin{cases} \psi(x) = A e^{-\kappa x} + B e^{\kappa x} & \text{for } x > a \\ \psi(x) = A_+ e^{ikx} + A_- e^{-ikx} & \text{for } |x| < a \\ \psi(x) = A' e^{-\kappa x} + B' e^{\kappa x} & \text{for } x < -a \end{cases}$$

Wavefunction must be normalizable

$$B = A' = 0$$

Wavefunction must be continuous everywhere

$$A e^{-\kappa a} = A_+ e^{ika} + A_- e^{-ika}$$

$$B' e^{-\kappa a} = A_+ e^{-ika} + A_- e^{ika}$$

$\frac{d\psi}{dx}$ must be continuous everywhere

(because $V(x)$ is finite everywhere)

$$-\kappa A e^{-\kappa a} = ik (A_+ e^{ika} - A_- e^{-ika})$$

$$+\kappa B' e^{-\kappa a} = ik (A_+ e^{-ika} - A_- e^{ika})$$

Therefore

$$\begin{cases} A_+ e^{ika} + A_- e^{-ika} = \frac{-ik}{\kappa} (A_+ e^{ika} - A_- e^{-ika}) \\ A_+ e^{-ika} + A_- e^{ika} = \frac{ik}{\kappa} (A_+ e^{-ika} - A_- e^{ika}) \end{cases}$$

$$\begin{pmatrix} 1 + \frac{ik}{\kappa} & (1 - \frac{ik}{\kappa})e^{-2ika} \\ (1 - \frac{ik}{\kappa})e^{-2ika} & 1 + \frac{ik}{\kappa} \end{pmatrix} \begin{pmatrix} A_+ \\ A_- \end{pmatrix} = 0$$

the determinant of this matrix must vanish in order to have a solution

$$\left(1 + \frac{ik}{\kappa}\right)^2 - \left((1 - \frac{ik}{\kappa})e^{-2ika}\right)^2 = 0$$

$$1 + \frac{ik}{\kappa} = \pm (1 - \frac{ik}{\kappa})e^{-2ika}$$

$$\left(1 + \frac{ik}{\kappa}\right)e^{ika} = \pm \left(1 - \frac{ik}{\kappa}\right)e^{-ika}$$

$$\left(1 + \frac{ik}{\kappa}\right)e^{ika}$$

$$= \left(\cos ka - \frac{k}{\kappa} \sin ka\right) + i \left(\sin ka + \frac{k}{\kappa} \cos ka\right)$$

must be purely real or purely imaginary

$$\sin ka = -\frac{\kappa}{k} \cos ka$$

$$A_+ = -A_-$$

$$A = e^{\kappa a} A_+ 2i \sin ka$$

$$B' = -e^{\kappa a} A_+ 2i \sin ka = -A$$

these are the odd solutions under $x \rightarrow -x$

$$\cos ka = \frac{\kappa}{k} \sin ka$$

$$A_+ = +A_-$$

$$A = e^{\kappa a} A_+ 2 \cos ka$$

$$B' = e^{\kappa a} A_+ 2 \cos ka = +A$$

these are the even solutions

$$k^2 + \kappa^2 = \frac{2m}{\hbar^2} (E + V_0 - E) = \frac{2mV_0}{\hbar^2}$$

$$V_0 \rightarrow \infty, \quad E \text{ fixed}$$

$$k \text{ fixed}, \quad K = \sqrt{2m(V_0 - E)} \frac{1}{\hbar} \rightarrow \infty$$

$$A \rightarrow 0, \quad B' \rightarrow 0$$

$$\bullet \text{ even solutions} \quad \cos ka = 0$$

$$ka = \frac{\pi}{2} (1 + 2m) \quad m \text{ integer}$$

$$\text{or } k = \frac{\pi}{L} m \quad (L = 2a) \quad m \text{ odd}$$

$$\bullet \text{ odd solutions} \quad \sin ka = 0$$

$$ka = m\pi \quad m \text{ integer}$$

$$\text{or } k = \frac{\pi n}{L} \quad \text{with } n \text{ even}$$

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$$\alpha = ka$$

$$\beta = \kappa a$$

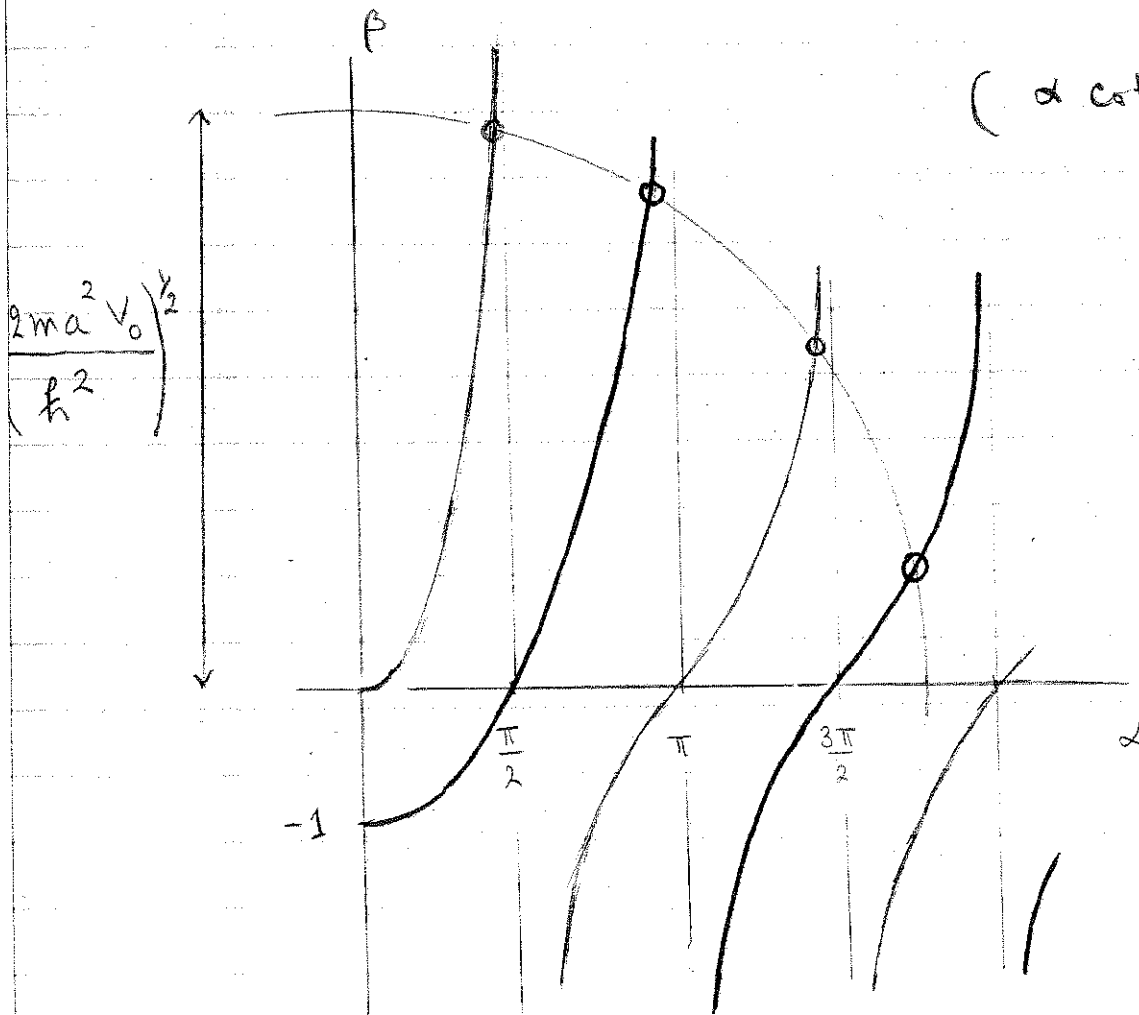
$$\alpha^2 + \beta^2 = 2ma^2 V_0 / \hbar^2$$

$$\beta = \alpha \tan \alpha$$

for even solutions

$$\beta = -\alpha \cot \alpha$$

for odd solutions



$$\left(\alpha \cot \alpha = \frac{\alpha \cos \alpha}{\sin \alpha} \rightarrow 1 \text{ as } \alpha \rightarrow 0 \right)$$

— odd

— even

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No matter how small $\frac{2ma^2V_0}{\hbar}$,

the first even solution always exists

The first odd solution stops existing

when

$$\left(\frac{2ma^2V_0}{\hbar^2} \right)^{\frac{1}{2}} \leq \frac{\pi}{2}$$

$$\text{or } V_0 \leq \frac{\pi^2 \hbar^2}{8ma^2}$$

$$\text{when } V_0 \rightarrow \frac{\pi^2 \hbar^2}{8ma^2},$$

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 \alpha^2}{2ma^2} \rightarrow \frac{\hbar^2}{2ma^2} \left(\frac{\pi}{2} \right)^2 = V_0$$